Mathematical Structure of Quantum Mechanics

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1. p-representation:

$$\left(\frac{p^2}{2m}+V(x)\right)|\psi\rangle = E|\psi\rangle$$

$$\Rightarrow \left(\frac{p^2}{2m}-E\right)\psi(p) = -\int \langle p|x'\rangle\langle x'|V(x)|x''\rangle\langle x''|p'\rangle\langle p'|\psi\rangle\, dx''dx'dp'$$

$$= +\frac{g}{2\pi\hbar}\int e^{-\frac{i}{\hbar}(p-p')x'}\cdot\delta(x'-a)\,\psi(p')\,dx'dp'$$

$$= \frac{g}{2\pi\hbar}\int e^{-\frac{i}{\hbar}(p-p')\cdot a}\cdot\psi(p')\,dp'$$

$$\Rightarrow \quad e^{+\frac{i}{\hbar}pa}\psi(p) = \frac{g/2\pi\hbar}{\frac{p^2}{2m}-E}\int e^{+\frac{i}{\hbar}p'a}\cdot\psi(p')\,dp'$$

$$\Rightarrow \text{(左右同時對 p 積分)}: \int e^{+\frac{i}{\hbar}pa}\psi(p)\,dp = \int \frac{g/2\pi\hbar}{\frac{p^2}{2m}-E}\,dp\cdot\int \psi(p')e^{+\frac{i}{\hbar}p'a}\,dp'$$

$$\therefore \int \frac{g/2\pi\hbar}{\frac{p^2}{2m}-E}\,dp = 1\ ,\ \text{let } \varepsilon=-E,\ \because E<0\ \therefore \varepsilon>0.$$

$$\Rightarrow \int \frac{dp}{p^2+2m\varepsilon} = \frac{\pi\hbar}{mg}$$

$$\Rightarrow \frac{1}{\sqrt{2m\varepsilon}}\tan^{-1}\left(\frac{p}{\sqrt{2m\varepsilon}}\right)\Big|_{p=-\infty}^{p=\infty} = \frac{1}{\sqrt{2m\varepsilon}}\cdot\pi = \frac{\pi\hbar}{mg} \Rightarrow \varepsilon = \frac{mg^2}{2\hbar^2} \Rightarrow E = -\frac{mg^2}{2\hbar^2}\ ※$$

$$\therefore \psi(p) = \frac{e^{ipa}}{\frac{p^2}{2m}+\frac{mg^2}{2\hbar^2}}\cdot N\ ,\ \Rightarrow \frac{1}{N^2} = \int|\psi(p)|^2dp = \int\frac{dp}{\left(\frac{p^2}{2m}+\frac{mg^2}{2\hbar^2}\right)^2} \Rightarrow \psi(p) = \sqrt{\frac{mg^3}{2\pi\hbar^3}}\cdot\frac{e^{ipa/\hbar}}{\frac{p^2}{2m}+\frac{mg^2}{2\hbar^2}}\ ※$$

($\hbar^2 \to \hbar^3$)

N: normalization constant

q-representation:

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E + g\delta(x-a))\psi = 0$$

积分 $\int_{a-\varepsilon}^{a+\varepsilon} dx,\ \varepsilon \to 0^+,$

则得 $\left.\frac{d\psi}{dx}\right|_{a+\varepsilon} - \left.\frac{d\psi}{dx}\right|_{a-\varepsilon} = -\frac{2mg}{\hbar^2}\psi(a)$ ——— ①

在 $x \neq a$ 之区域, $\frac{d^2\psi}{dx^2} + \frac{2mE}{\hbar^2}\psi = 0$

$$\text{let } \beta = \sqrt{-\frac{2mE}{\hbar^2}}, \quad (\because E < 0)$$

$$\Rightarrow \frac{d^2\psi}{dx^2} + (-\beta^2)\psi = 0 \Rightarrow \psi = Ae^{\beta(x-a)} + Be^{-\beta(x-a)}$$

$\because \psi$ 以 a 為中心左右對稱, 且 $\psi \to 0$ as $x \to \infty$ or $-\infty$.

$$\therefore \begin{cases} \psi = ce^{-\beta(x-a)}, & \text{for } x > a \\ \psi = ce^{+\beta(x-a)}, & \text{for } x < a \end{cases} \Rightarrow \begin{cases} \psi' = -\beta c e^{-\beta(x-a)}, & x > a \\ \psi' = \beta c e^{+\beta(x-a)}, & x < a \end{cases}$$

代入①式, $-2\beta c = -\frac{2mg}{\hbar^2}c$

$$\Rightarrow -\frac{2mE}{\hbar^2} = \frac{m^2g^2}{\hbar^4} \Rightarrow E = \frac{-mg^2}{2\hbar^2} \quad ※$$

$$\Rightarrow \psi(x) = \frac{\sqrt{mg}}{\hbar} e^{-\frac{|x-a|\cdot mg}{\hbar^2}} \quad ※$$

/0

2. quantum state $|\psi\rangle$ must satisfy

① $(p-\langle p\rangle)|\psi\rangle = c(x-\langle x\rangle)|\psi\rangle$

② $\langle\psi|(p-\langle p\rangle)(x-\langle x\rangle)+(x-\langle x\rangle)(p-\langle p\rangle)|\psi\rangle = 0$

① in p-representation:

$$(p-\langle p\rangle)\psi(p) = c\cdot\left(i\hbar\frac{d}{dp}-\langle x\rangle\right)\psi(p)$$

$$\Rightarrow (p-\langle p\rangle+c\langle x\rangle)\psi(p) = i\hbar c\cdot\frac{d\psi(p)}{dp}$$

$$\Rightarrow \frac{d\psi}{\psi} = \frac{i}{\hbar}\left(\frac{p-\langle p\rangle}{c}+\langle x\rangle\right)dp$$

we can shift our origin $\langle p\rangle = 0$, i.e, in a new frame of reference $\langle p\rangle = 0$.

$$\therefore \frac{d\psi}{\psi} = \frac{i}{\hbar c}(p+c\langle x\rangle)dp$$

$$\Rightarrow \psi = \psi(0)\cdot e^{i\langle x\rangle p/\hbar}\cdot e^{ip^2/2\hbar c}$$

$\because (p-\langle p\rangle)|\psi\rangle = c(x-\langle x\rangle)|\psi\rangle$ and $\langle\psi|(p-\langle p\rangle) = \langle\psi|(x-\langle x\rangle)c^*$,

② becomes: $\langle\psi|\frac{1}{c}p^2+(\frac{1}{c})^*p^2|\psi\rangle = 0$

$$\Rightarrow \langle\psi|p^2|\psi\rangle\cdot\left(\frac{1}{c}+(\frac{1}{c})^*\right) = 0$$

So that $\frac{1}{c}$ must be pure imaginary, let $\frac{1}{c} = i\alpha$, α is real. ※

$$\therefore \psi = \psi(0)\cdot e^{i\langle x\rangle p/\hbar}\cdot e^{-p^2\alpha/2\hbar}$$

Shift the origin back. Then we have:

$$\Psi(p) = \Psi(\langle p \rangle)\, e^{i\langle x \rangle (p-\langle p \rangle)/\hbar} \cdot e^{-(p-\langle p \rangle)^2/\Delta}, \quad \text{where } \Delta = \frac{2\hbar}{\alpha}$$